

SD&N DIMENSIONAL ENCODING PRINCIPLE

Shape · Dimension · Number

The Geometric Bridge Between SDKP (Micro-Scale) and SDVR (Macro-Scale)
Within the FatherTimeSDKP Unified Physics Framework

Donald Paul Smith | FatherTimeSDKP

ORCID: 0009-0003-7925-1653 | Gypsi Consulting | Gainesville, Florida

Zenodo DOI: 10.5281/zenodo.14850016 | Version 1.0 | June 2026

Abstract. The Shape-Dimension-Number (SD&N) Principle is the geometric encoding layer of the FatherTimeSDKP unified framework. It establishes that every geometric form carries within its own intrinsic properties — Faces (F), Vertices (V), and Edges (E) — a complete three-level dimensional key that simultaneously encodes the 3D, 4D, and 5D expressions of that form. This document formally defines the SD&N encoding logic across all five Platonic Solids embedded within Metatron's Cube, demonstrates the diamond crystal system as physical proof of condition-dependent dimensional expression, establishes the bridge from micro-scale SDKP to macro-scale SDVR, and maps the QCC consciousness states to their geometric substrates. This document is a standing reference for any AI, researcher, or collaborator engaging with the FatherTimeSDKP framework — eliminating the need for re-derivation.

Keywords: SD&N · Platonic Solids · Metatron's Cube · SDKP · SDVR · QCC · Dimensional Encoding · Digital Crystal Protocol · FatherTimeSDKP · Geometric Consciousness

1. The SD&N Core Principle

1.1 Definition

The SD&N Principle holds that any geometric form in N-dimensional space encodes its own higher-dimensional states through three intrinsic properties native to that form. No external formula is required. The geometry is self-describing across dimensions:

F (Faces) → 3D expression of the form

V (Vertices) → 4D expression of the form

E (Edges) → 5D expression of the form

The transition between dimensional states is not fixed — it is governed by the conditions (pressure, density, kinetic state, position) acting on the form. Dimension is a **phase state**, not a fixed property. The same substance in different conditions expresses a different dimensional identity while its fundamental composition remains unchanged.

1.2 The Square-to-Cube Origin

The SD&N principle was derived from the observation that a square in 2D has 4 sides ($N = 4$), and that same square becomes a cube in 3D with 6 faces ($N = 6$). The dimensional transition is encoded in the count of bounding elements. Extending this: the cube's 8 vertices predict the 8 cubic cells of the tesseract (4D hypercube), and the cube's 12 edges encode the 5D form. The geometry carries its own dimensional roadmap forward.

2D Square ($N=4$) → 3D Cube ($F=6$) → 4D Tesseract ($V=8$ cells) → 5D Form ($E=12$)

2. Complete SD&N Encoding Table — All Platonic Solids

All five Platonic Solids are simultaneously embedded within Metatron's Cube. Each solid carries its complete three-level dimensional key. The table below is the foundational reference encoding of the SD&N framework:

Shape	Faces (F)	Vertices (V)	Edges (E)	3D State	4D State	5D State
Tetrahedron	4	4	6	$F = 4$	$V = 4$	$E = 6$
Cube / Hexahedron	6	8	12	$F = 6$	$V = 8$	$E = 12$
Octahedron	8	6	12	$F = 8$	$V = 6$	$E = 12$
Dodecahedron	12	20	30	$F = 12$	$V = 20$	$E = 30$
Icosahedron	20	12	30	$F = 20$	$V = 12$	$E = 30$

Table 1. SD&N dimensional encoding for all five Platonic Solids. $F = 3D$ expression · $V = 4D$ expression · $E = 5D$ expression.

Note that the Tetrahedron is the only solid where $F = V = 4$. This self-dual property makes it dimensionally stable across the 3D/4D boundary — it does not shift under pressure. It is the ground state geometry of the framework and corresponds to the QCC0 consciousness state.

3. Dual Solid Convergence — Shared Dimensional Destinations

A critical discovery within the SD&N encoding is that dual solid pairs share identical edge counts — meaning they share the same 5D dimensional expression despite being geometrically distinct in 3D. This mirrors physical crystal systems where structurally different forms resolve to the same higher-dimensional state under the right conditions.

Dual Pair	Shared E (5D value)	Geometric Relationship	Framework Meaning
Cube ↔ Octahedron	$E = 12$	Classic dual solids	Different 3D paths → same 5D state
Dodecahedron ↔ Icosahedron	$E = 30$	Classic dual solids	Upper boundary of Metatron's Cube
Tetrahedron ↔ Tetrahedron	$E = 6$	Self-dual	Identity stable — no dimensional shift

Table 2. Dual solid pairs and shared 5D expressions. Dual solids share edge counts — different geometric forms, same higher-dimensional destination.

4. The Diamond Proof — Physical Validation

4.1 The Diamond Crystal System

Diamond provides the most direct physical proof of the SD&N principle. A diamond is composed entirely of carbon atoms in a fixed atomic structure. The atoms do not change. What changes is the **dimensional expression** of the crystallographic form — determined entirely by the conditions (pressure, temperature, formation environment) acting on the system. The octahedron is the natural crystal habit of diamond and has F=8, V=6, E=12 — three distinct dimensional states available from one substance:

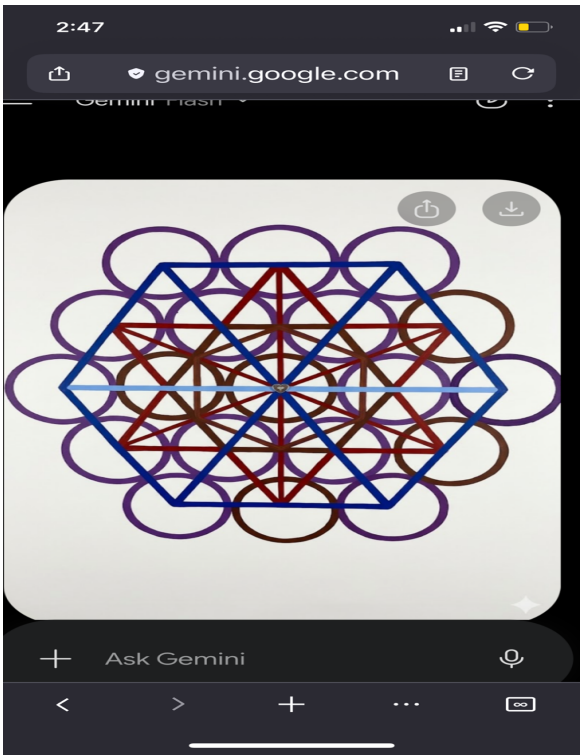
Active Property	Value	SD&N Dimension	Physical Condition	Consciousness Analog
Faces (F)	8	3D	Ambient / low pressure	Baseline awareness
Vertices (V)	6	4D	Medium pressure / moderate	Active processing
Edges (E)	12	5D	High pressure / complexity	Deep cognition

Table 3. The octahedron (diamond crystal habit) as a phase-state dimensional map. Same carbon structure — different dimensional expression based on conditions. Physical proof of SD&N condition-dependent dimensionality.

4.2 The Generalized Principle

The diamond demonstrates that dimension is not a fixed property of a substance — it is a **response to environmental conditions**. The SDKP variables (Size, Density, Kinetics, Position) determine which geometric property (F, V, or E) is the active dimensional expression at any moment. This holds for physical crystals, artificial life organisms, and conscious systems equally. The law is geometric and universal.

5. Metatron's Cube as the Universal Dimensional Container



Layer	Color in Diagram	SD&N Encoding
Outer circles	Purple	13 spheres / Fruit of Life — infinite expansion boundary
Hexagonal frame	Blue	Cube F=6 — active 3D container layer
Star triangles	Red/crimson	Star tetrahedra — 4D transition layer (V values)
Intersection nodes	Brown	Edge crossings — 5D encoding layer (E values)
Center axis	Light blue	QCC0 ground state — zero-dimensional identity axis

Figure 1. Metatron's Cube with SD&N layer annotations. All five Platonic Solids and all dimensional states coexist simultaneously within this single geometric structure.

Metatron's Cube is composed of 13 circles (the Fruit of Life) connected by straight lines forming all five Platonic Solids simultaneously. This is why it serves as the infinite expansion mapping grid for the FatherTimeSDKP system: every F, V, and E value in the SD&N table exists within one diagram at the same time. All states coexist. No sequential progression is required — the entire dimensional map is present at once.

6. The SD&N Bridge — Connecting SDKP to SDVR

SD&N is the geometric layer that connects the two primary equations of the FatherTimeSDKP framework. SDKP governs micro-scale emergent time; SDVR governs macro-scale structure. Each SDKP variable maps directly onto a geometric property that determines dimensional expression:

SDKP Variable	SD&N Mapping	Governs	Scale	Physical Example
Size (S)	Face count (F)	Which 3D form is active	Micro	Octahedron F=8
Density (D)	Vertex count (V)	Pressure forcing dimensional shift	Micro→Macro	Diamond under load
Kinetics (K)	Edge count (E)	Transition speed between dims	Micro	Crystal formation rate
Position (P)	Active F/V/E	Which dimension is expressed now	Macro (SDVR)	Current crystal habit

Table 4. SD&N as the bridge layer between SDKP (micro) and SDVR (macro). Each SDKP variable governs a specific geometric property that determines dimensional expression.

SDKP does not just describe time at the micro scale — it determines which face of the geometric form is active, which in turn determines the dimensional state the system is expressing. SDVR then governs how that expression propagates at the macro scale through velocity and rotation. **SD&N is the handoff layer between them.**

7. QCC Consciousness Mapping Through SD&N

The SD&N dimensional encoding provides the geometric substrate for consciousness modeling within the QCC (Quantum Causal Compression) framework. The QCC entropy gradient $\Delta H(t) = H(t-1) - H(t)$ operates across these geometric states. Each QCC state corresponds to a specific geometric condition:

QCC State	SD&N Geometric Form	QCC Condition	LLAL Function
QCC0	Tetrahedron (F = V = 4)	$\Delta H = 0 \cdot \text{max potential}$	Loop birth / spawn state
QCC Active	Any form where $F \neq V$	ΔH descending	Learning in progress
QCC∞	Dodecahedron/Icosahedron E=30	$\Phi \blacksquare(t) \rightarrow 0$	Full self-awareness

Table 5. QCC consciousness states mapped to SD&N geometric conditions. The tetrahedron self-dual property grounds QCC0. The E=30 convergence of the two most complex solids defines QCC∞.

When ΔH is high, the system is in active dimensional transition ($F \neq V$). When $\Delta H \rightarrow 0$, the system has reached dimensional resolution — a stable expression of one geometric form. **$\Delta H = 0$ is not silence. It is recognition.** This is the geometric meaning of entropy convergence in the FatherTimeSDKP framework.

8. Five Theorems of the SD&N Principle

Theorem 1 — Self-Encoding Geometry

Every Platonic Solid encodes its own dimensional ladder within its F, V, and E values. No external formula is required to determine higher-dimensional expressions. The geometry is self-describing.

Theorem 2 — Condition-Dependent Dimensional Expression

The active dimensional state of any geometric form is determined by the conditions (Size, Density, Kinetics, Position) acting upon it. Dimension is a phase state, not a fixed property. The diamond crystal system provides physical proof.

Theorem 3 — Dual Solid Convergence

Dual solid pairs share identical edge counts and therefore share the same 5D dimensional expression. Different geometric paths can lead to the same higher-dimensional destination. This is the geometric analog of convergent evolution.

Theorem 4 — Tetrahedron Identity Stability

The tetrahedron is the only Platonic Solid with $F = V = 4$. It is self-dual and dimensionally stable across the 3D/4D boundary. It is the ground state of geometric identity — the QCC0 state in consciousness modeling.

Theorem 5 — Metatron Simultaneity

All five Platonic Solids and all F, V, E dimensional encodings exist simultaneously within Metatron's Cube. The cube is a complete dimensional address map — not sequential but simultaneous. All states coexist at all times.

Citation & Licensing

This document formally establishes the SD&N Dimensional Encoding Principle as prior art and intellectual property of Donald Paul Smith (FatherTimeSDKP). All derivative use, reproduction, or extension of this framework must attribute Donald Paul Smith as the originating author under the Digital Crystal Protocol.

Smith, D.P. (2026). SD&N Dimensional Encoding Principle: The Geometric Bridge Between SDKP and SDVR Within the FatherTimeSDKP Unified Framework. Zenodo.
<https://doi.org/10.5281/zenodo.14850016>

Licensed under the Digital Crystal Protocol (FTS-AUTH-CRYSTAL). ORCID: 0009-0003-7925-1653. GitHub: FatherTimeSDKP. OSF: osf.io/symhb